

DATA-DRIVEN TEA AUCTION PRICE PREDICTION FOR THE COLOMBO TEA AUCTION

Project ID: 25-26J-292

Project Proposal Report

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BSc (Hons) in Degree in Information Technology
Specialising in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

The paper explores the work of creating an automated machine-learning model to predict the price of the tea auction in Sri Lanka with the aim of replacing the current manual model-based price prediction process maintained by the Sri Lanka Tea Board. Through the years, the Colombo Tea Auction, opened on 30 July 1883 when one lot of the five lots on offer was sold, has grown into the world leader in the volume of lots sold, currently moving around 12,000 in a week schedule, with three auction rooms. The research thus relates to the critical necessity of automating the complicated manual process of price forecasting done by Sri Lanka Tea Board professionals. Despite the fact that the conventional manual methods rely on vast amounts of expert knowledge, they find significant constraints when applied to the multitude of variables that are to be processed at one and the same time. The research aims will involve designing and developing a smart automated model of forecasting that might handle the large variables that are already being manually evaluated by tea-board experts- quality grades, moisture content, seasonal variations, tea-grade descriptions (OP, BOP, BOPF, Pekoe, etc.), altitude-based divisions (low-grown: sea-level-600 m, mid grown 600 1200 m, high grown above 1200 m), leaf-size specification, price trend over the years, broker recommendations, and the prevailing situation. The system is expected to provide accurate automated comments that guide buyer and other stakeholders in making decisions about the best way of tea purchases and help the tea board expand their price prediction capacity according to the large operation requirements needed by the auction. XGBoost will be used as the original value model of the validation study comparing the prediction accuracy with the proposed ensemble model with the actual overall nature of the auction price data.

Keywords: Automated tea price prediction, Machine learning automation, Buyer decision support system, Ceylon tea

Table of Contents

<i>Declaration</i>	<i>iii</i>
<i>Abstract</i>	<i>iv</i>
<i>List of Figures</i>	<i>vii</i>
<i>List of Tables</i>	<i>vii</i>
<i>1. INTRODUCTION</i>	<i>viii</i>
1.1 Background & Literature survey	X
1.1.1 Historical Context of Tea Auction Systems	X
1.1.2 Tea Quality Assessment and Pricing Mechanisms	X
1.1.3 Existing Statistical Approaches to Price Prediction	X
1.1.4 Machine Learning Applications in Agricultural Commodity Pricing.....	xi
1.1.5 Digital Auction Platforms and Data Availability	xi
1.2 Research Gap	xii
1.3 Research Problem	xiii
1.3.1 Problem Statement	xiii
1.3.2 Inconsistency in Manual Evaluation	xiii
1.3.3 Scalability Crisis	xiii
1.3.4 Resource Intensity and Expertise Dependency	xiv
1.3.5 Research Questions	xiv
1.3.6 Hypothesis	xiv
2. Objectives	xv
2.1 Main Objective	xv
2.2 Specific Objectives	xv
2.2.1 Automated Process Development	xv
2.2.2 Machine Learning Model Architecture	xv
2.2.3 Validation Against Manual Predictions and Benchmarks	xvi
2.2.4 Decision Support System Development	xvi
2.2.5 System Validation and Tea Board Integration.....	xvi
3. Methodology	xvii
3.1 System Architecture and Components	xvii
3.2 Technology Stack and Implementation Approach	xviii
3.3 Validation Framework and Performance Evaluation	xix
3.4 Design	xx
3.5 Development and Implementation	xx
3.6 Use Case Diagram	xxi
3.7 Work Breakdown Structure	xxii
3.8 Gantt Chart	xxii
3.9 Project Requirements	xxiii
3.9.1 System Requirements.....	xxiii
3.9.2 User Requirements	xxiii
4. Description of Personal and Facilities	xxiv

5. Budget and Budget Justification	xxv
6. References.....	xxvi

List of Figures

Figure 1 System Comparison	xvii
Figure 2 Data-Driven Tea Auction Price Prediction Use Case Diagram	xxi
Figure 3 Data-Driven Tea Auction Price Prediction Use Case Diagram	xxii
Figure 4 Data-Driven Tea Auction Price Prediction Gantt Chart.....	xxii

List of Tables

Table 1System Comparison.....	xii
Table 2 System Comparison.....	xxv

1. INTRODUCTION

The tea industry to Sri Lanka is one of the major economic lines that have brought in an approximate sum of US\$1.2 billion and sustaining about 2.1 million livelihoods every year [9]. Since the first tea bush was planted by James Taylor in 1867 at Kandy, the industry has come a long way giving the humble beginning moving towards cultivation of tea as a global powerhouse in terms of tea production and producing some of the finest tea in the world across 7 distinct agro-climatic regions [4], [11].

It has had a substantive influence over Ceylon tea prices due in part to it having been the main price discovery mechanism at the Colombo Tea Auction which has operated since 1883 [4]. It has since risen to become the biggest tea auction by number of lots in the world handling as many as 12,000 lots a week in three auction halls. However, as set out in the larger scale due to the catalysing effect of the COVID-19 pandemic that accelerated the digital transformation in the month of April 2020, auction price determination is a completely manual process that limits the ability to effectively make decisions in real-time [1], [3].

The aim of the current study is to develop an intelligent automated forecasting system that would be able to execute the comprehensive set of variables currently being considered by the professional panel appointed by the tea board [2]. Such factors consist of quality grades, moisture content, seasonality, the different tea kinds of grades (OP, BOP, BOPF, Pekoe, etc.), altitude categories (low-grown: sea level-600 m, mid-grown: 600 m-1200 m, high-grown: above 1200 m), leaf-size grades, old prices, and broker quotes, and the state of current supply [2], [5], [7]. The proposed system allows making highly precise forecasts and is therefore capable of improving the level of decision-making on the part of the buyers and other market members, helping to devise the most favourable purchasing strategies. At the same time, it also enables the tea board to engage in wider range of price predictions activities comparable to the extensive operational needs of the auction [6].

The direct effect on the various stakeholders to ensure informed decision-making is that buyers could plan their purchase during the optimal point based on the real automated prediction, producers could plan their production and time to sell, brokers could offer to do more advisory services, and the tea board could match the volume of its operation to the carrying out of auctions [12].

1.1 Background & Literature survey

1.1.1 Historical Context of Tea Auction Systems

History of the evolution of the Colombo Tea Auction is connected with the broad changes in the global commodity markets [4], [10]. As advances, which began in the 1990s with the replacement of manual outcry procedures with digitized records-keeping and, most recently, with the further evolution of digitized platforms to fully digital, have increased the possibilities of data gathering and interpretation. The digital age, in its turn, assists in the systematic recording of bidding behaviours, temporal pricing patterns and quality-price correlations that were previously observable only anecdotally [3].

1.1.2 Tea Quality Assessment and Pricing Mechanisms

There is persistent research that has been conducted into the relationship between certain quality specifications and the auction prices. The moisture level, the type of grade, and origin region as the key factors that influence the price, although the seasonality and market sentiment were indicated as secondary price determinants. However, the use of correlation analysis and linear-regression formats were major methodologies adopted in these studies and thus restricted possibilities relating to perceiving nonlinear relationships or interaction measures [2], [5].

1.1.3 Existing Statistical Approaches to Price Prediction

Early projects of tea price forecasting in literature demanded the use of classical time-series methodologies by the scholars. Box-Jenkins ARIMA models were one of the early natural occurring attempts to do so; Fernando and Silva (2019) give Mean Absolute Percentage Errors (MAPE) of 0.26-6.03% of the BOPF grade on various elevation settings. Based on the findings, the SARIMAX extension with external variables including supply, demand, and rainfall, produced only modest increases in accuracy of predictions [2].

Vector Autoregression (VAR) has been used in more advanced models to pick up dependence between tea grades, and regions. The study by Perera et al. (2020)

indicates that the effect of regional price spillovers is a notable factor that determines the pricing of individual lots especially when there are quality seasons associated with monsoon transitions.

Decomposition in seasons is also another useful method in terms of cyclical trends. The ability to identify quality seasons has aided in the development of more precise seasonal forecasting models since these are the seasons when specific regions perform better in terms of making superior teas due to preferred weather conditions (Jayasinghe and Rajapakse, 2018) [9].

1.1.4 Machine Learning Applications in Agricultural Commodity Pricing

Over the past decade, research studies on machine learning have heightened with respect to price formation of agricultural commodities. Support vector machines (SVM) have proved to be superior in situations involving the prediction of non-linear scenarios especially with the application of several feature dimensions. The neural network techniques such as Multi-Layer Perceptrons (MLP) and Long Short-Term Memory (LSTM) networks have ever provided better performance in time-series prediction situations where a complex temporal distance is observed. Convolutional Neural Networks (CNN) have also been explored on the integration of the data on image-based quality assessments into pricing models [5].

1.1.5 Digital Auction Platforms and Data Availability

The development of online auction platforms has produced a new set of data which can be subject to complex analytic tools. Machine learning models get an extensive set of features to learn with real-time bidding data, buyer behaviour, historical auction records, meteorological data and automated quality assessment. However the collection of disparate sources of data such as historical auction records, weather statistics, quality evaluation and market indicators pose significant preprocessing challenges.

1.2 Research Gap

Table 1 System Comparison

Aspect	Existing Systems (Research)	Proposed Application Component
Machine Learning Techniques	Traditional statistical methods with limited exploration of modern ML approaches.	Advanced ensemble methods, deep learning architectures, and attention-based models specifically optimized for tea price prediction.
Data Integration	Single data type focus pricing or quality or weather with limited cross-domain correlation.	Multi-modal data fusion of real-time bidding behaviors, automated quality scoring, local climatic conditions and worldwide market signals.
Regional Coverage	Single tea grade (BOPF) or limited regional analysis without comprehensive geographical modeling	Coverage more of Sri Lanka's agro-climatic regions with distinct quality profiles and pricing pattern recognition
Validation Framework	Academic validation without industry benchmark comparison or standardized evaluation metrics	Industry-standard validation against established benchmarks with practical forecasting method comparisons

1.3 Research Problem

The Sri Lankan tea industry is now faced by a series of critical challenges to scale and optimise a manual price-prediction process operated by the Tea Board [4]. Despite the historical dependency on higher level human judgment in this process, the disproportionate rise in auction volumes due to increased numbers of lots offered each week (12,000+) has produced some operational and customer-oriented strategic challenges that are vehemently demanded to be addressed by automating options [1], [6].

1.3.1 Problem Statement

The existing manual models of tea price forecast exhibit several critical limitations, which have been strongholds to market efficiency and limits to various stakeholder decision-making.

1.3.2 Inconsistency in Manual Evaluation

Artificiality of the manual price prediction due to subjectivity by human causes divergence in the price between similar lots, hence making them inefficient in the market and less predictable. Because of the complexity of having to consider multiple variables, such as the grades of quality, moisture, seasonality, elevation, size and historical trends, as well as input by brokers and the conditions of supply, the manual optimization of the same is extremely challenging.

1.3.3 Scalability Crisis

The manual evaluation process of the Tea Board cannot be scaled at the level that is demanded by the modern market where 12,000 or more lots are auctioned at a single week in seven agro-climatic zones [4], [6]. Manual assessment of each lot's quality grades, moisture content, seasonal variations, grade classifications, elevation factors, size specifications, historical patterns, broker valuations, and supply conditions can't handle.

1.3.4 Resource Intensity and Expertise Dependency

The current system requires significant human resources and specialized expertise which can not easily grow along with the increasing volumes of auctions or be available at all evaluation situations.

1.3.5 Research Questions

1. Are machine learning algorithms able to efficiently automate the complex manual price prediction process currently undertaken by Tea Board experts, and improve or at least equal prediction accuracy?
2. How do computer systems most effectively integrate and process the entire array of variables currently being measured manually (quality grades, moisture content, seasonal fluctuation, grade classes, elevation, size, trend history, broker valuations, supply conditions)?
3. How can automated price forecasts be structured to best support buyers and stakeholders in making optimal tea purchasing strategies?
4. How do computerized ML predictions compare with current manual Tea Board predictions and XGBoost benchmarks regarding accuracy, consistency, and processing efficiency?
5. Can the automated system provide real-time price predictions that enable real-time decision-making during auctioning processes?

1.3.6 Hypothesis

The research is guided by the following primary hypothesis.

“The application of more sophisticated machine-learning algorithms is shown to automate the currently manual task of predicting tea prices conducted by the experts of the Sri Lanka Tea Board at an overall higher degree of accuracy, consistency, and scalability. Moreover, the methods provide real-time decision assistance to the buyers and other stakeholders that are engaged during the establishment of ideal tea-buying plans.”

2. Objectives

2.1 Main Objective

In devising and validating an automated machine learning system that can replicate and further perfect the current manual processes of tea price prediction that is currently being conducted by the Sri Lanka Tea board, this study aims to solve this problem. With such a system buyers and other stakeholders would get real-time data-based decision support as they develop the best purchasing strategies [3], [12].

2.2 Specific Objectives

2.2.1 Automated Process Development

Automated price prediction processes have to be developed to substitute the current manual price prediction processes conducted by Tea Board experts. The major goal would be to develop entire ingestion systems that will be able to absorb all of the variations that are currently being handled manually like quality grades, moisture content, seasonal fluctuations, grade variants, elevation levels, size measures, historical trends, valuation by the broker, and compartmentalization of supplies. More importantly, in this project, there must be the development of and addressing feature engineering frameworks to become an embodiment of the rich decision-making logic incorporated in the manual evaluation processes.

2.2.2 Machine Learning Model Architecture

This project aims at developing innovative machine-learning frameworks that can integrate a combination of heterogeneous variables and evaluate them collectively, echoing integrated assessment procedure that is used by Tea Board specialists. The method implies the construction of specially designed algorithms on the distribution of variables of categorical types (types of degrees, altitude, place of origin) and opposite structures on the distribution of variables of continuous nature (moisture, historical price curve, aggregate supply curve). A combination structure is then implemented to combine the respective merits of these singular predictors and reconstruct the overall judgemental observations that are attained by its human-experts.

2.2.3 Validation Against Manual Predictions and Benchmarks

An evaluation experiment was performed to check the abilities of automated machine-learning (ML) predictions against manual predictions made by The Tea Board of India with regards to tea leaf grade estimations [2], [5]. An XGBoost model was trained using the same input variables with which the board used, and became the basis of comparison in this comparative analysis. Rigorous testing was performed against live data of actual auction prices and the automated method should have at least as good a pass rate as the manual method.

2.2.4 Decision Support System Development

Among the areas to be developed more carefully within the scope of Decision Support Systems, one should distinguish the process of introducing them. Some of the key goals include the establishment of user friendly interfaces that can present automated predictions in forms compatible with the decision process of the buyers. Also, recommendation systems must be developed sharing buyers with a decision on choosing the teas based on their predicted prices and estimates of their quality. In addition, alert systems should be implemented in order to see the best commercial opportunities based on automated analyses.

2.2.5 System Validation and Tea Board Integration

Conduct comprehensive testing compare automated prediction of the quality of tea with manual assessments provided by the Tea Board. Validate system performance across varied tea grades, geographical origin and market conditions a practice that provided a solid validation measure.

3. Methodology

Prior studies allow one to place the current investigation in the concept of a mixed-methods paradigm of collecting, processing, and analyzing data when both quantitative machine-learning algorithms and qualitative expert judgments will be used. The research employs a controlled developmental context Systems Development Lifecycle which is laced with agile environments to enable an iterative improvement and strict evaluation with the traditional manual prediction procedures of the Tea Board [14], [15].

3.1 System Architecture and Components

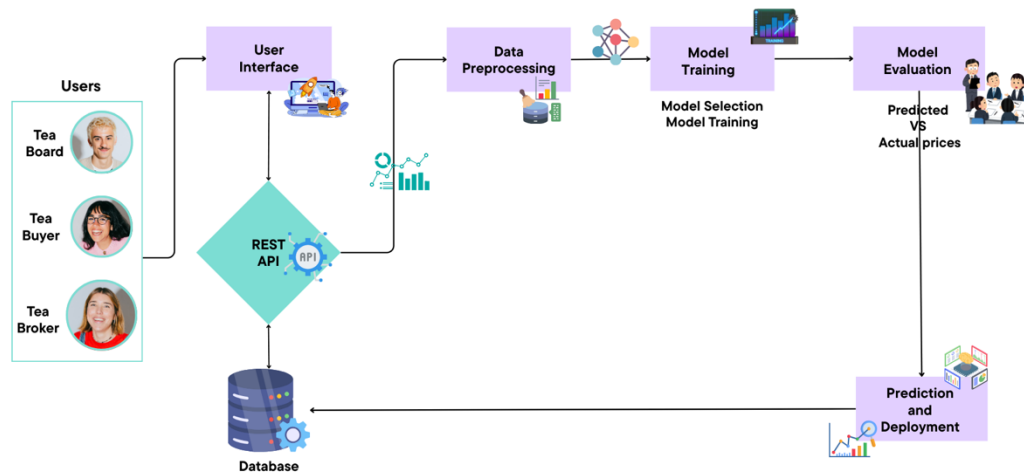


Figure 1 System Comparison

Methodological guidelines entail a step-by-step, progressively oriented process that propounds the adaptation as well as streamlining of intrinsic operations of an automated tea auction prediction (ATAP) design. The Historical Data Ingestion Module is the main component of data collection and processes Colombo Tea Auction data over 2015-2024. The module drives data continuity, data integrity in case of heterogeneous auction formats and covers several stages of digital transformation. The Real-time Data Preprocessing Engine can therefore preprocess raw auction data by performing operations that handle missing data, outliers and incompatibilities amid grade classifications, regions of origin, or quality rating; only uniquely high quality

records can be forwarded onto the next processes. Tea-industry specific feature engineering methods are utilized by the Automated Feature Extraction and Transformation Pipeline to extract the salient variables such as the quality grades, region-specific characteristics, and seasonal patterns, and a broader market dynamic that further improve predictive performance. The Automated Model Training and Validation Module uses ensemble methods: XGBoost, Random Forest, Support Vector Machine, and Deep Neural Network to cover the naturally high complexity of tea auction pricing; the algorithms used get dynamically parameterised to reach maximum prediction accuracy. Lastly, the Real-time Prediction and Recommendation System reads auction lot data, applies trained ensemble models to provide price predictions and recommendations and is designed to accept over 12 000 lots a week, while processing them with less than two seconds per prediction.

3.2 Technology Stack and Implementation Approach

The current investigation uses packages of advanced technologies and procedures in consecutive steps of a Tea Auction Prediction System. Python is used as the main programming language with the flow of data preprocessing pipelines, modules of feature extraction, and implementation of machine learning models. The most important libraries such as NumPy, pandas, Scikit-learn, XGBoost, and TensorFlow are used to facilitate a massive manipulation and sophisticated analytical tasks appropriate to analyzing the real-time data in an auction. The system architecture is inherently layered: the Data Collection Layer deals with the old auction databases and real time auction feeds that constantly watch market parameters, the Data Processing Layer deals with preprocessing engines and feature transformation pipelines where auction data is cleansed and prepared to become the input of machine learning models, Model Training Layer contains the ensemble machine learning algorithms that will receive the prepared data and then generate predictive models, and finally the Prediction Layer aims to offer real time price forecasting and recommendations to Tea Board professionals and other stakeholders of the auction.

3.3 Validation Framework and Performance Evaluation

Validation is also an important part of the auction price prediction system proposed and it is important to ensure that forecasts generated by the system can be accurate and reliable so that they become usable by the Tea Board in a practical application. The quantitative framework is historically validated, requiring that the predictions would be able to perform better than the actual auction results through statistical significance testing as well as several established measures, with a Mean Absolute Percentage Error (MAPE) of under 5 % in the case of main tea grades, a Root Mean Square Error (RMSE) and R-squared values in excess of 0.85 indicating excellent explanatory power. Out-of-sample testing is used to demonstrate the generalizing information of this model in different time frames and market situations and expert validation entails objective evaluation of Tea Board professionals who compare the automated recommendations with conventional manual predicting methods. Technical scores covered in the performance of the system will be based on the technical metrics such as the accuracy levels in various tea grades and regions, the speed of data processing at over 12000 lots per week during the auctions, the responsiveness of the system (real time) which allows the system to generate predictions in 2 seconds and the reliability as the system will need to have a 99 % uptime during auction periods. Success criteria consist of quantitative standards that involve statistical significance in predictive improvements over current manual ones as well as qualitative standards including expert evaluation scores over 4.0 out of a 5.0 scale of domain professionals as well as stakeholder adoption levels above the 80 % level among auction participants, and a complete exercise of user acceptance testing that confirms practical usefulness of the system to the Tea Board operations and auction decision processes.

3.4 Design

At this point, the tea auction price prediction system has been broken down to four fundamental components of architecture as part of ensuring a systematic development in the project time frame. The Analytics Engine involves pipelines of feature engineering which manipulates the data on tea quality parameters, seasonal trends, and market signals into model-ready data. The Prediction Core is an ensemble machine learning model specifically trained at historical auction data with the ability to make multi-grade price predictions. Lastly, the User Interface Layer offers sensible dashboards to both brokers and traders with visualizations and confidence measurement of the predictions. The server side involves the presence of cloud-based GPU clusters that support the increment of training, secured databases that support sensitive auction data and streaming of real-time data gateways. The architecture of the system is designed to be scalable enable it to support an influx during the auction turn and modular to add on future features such as global market integration.

3.5 Development and Implementation

The present work amounts to the creation of the data preprocessing pipeline that would be needed to handle three salient challenges of the tea auction dataset, namely, inconsistency of the grade labels, limited meteorology metrics, and time-dependent heterogeneity of the output data. The next steps include the application of the ensemble learning algorithms the gradient boosting and neural networks in the cross-validation technique using the past history of occurrences of many sources and seasons. The models coming out of it offer medium between interpretability and predictive accuracy thus allowing brokers and other interested parties to justify the strategic decisions. In the operation testing stage, we will incorporate information on live auction feeds into the trained models, hence providing real-time, price and authentication signals as well as various access privileges within each user group. Back testing evaluation will be done using hybrid backtesting procedures, wherein whereby the predictions of the model will be compared to the historical auction prediction. This system will be launched on a one-time simulated auction cycle and subsequent feedback cycle will observe the accuracy of the predictions, the latency and receive conversion on user

feedback to assure the industry standard reliability and usability in the extremely dynamic auction area [3], [7].

3.6 Use Case Diagram

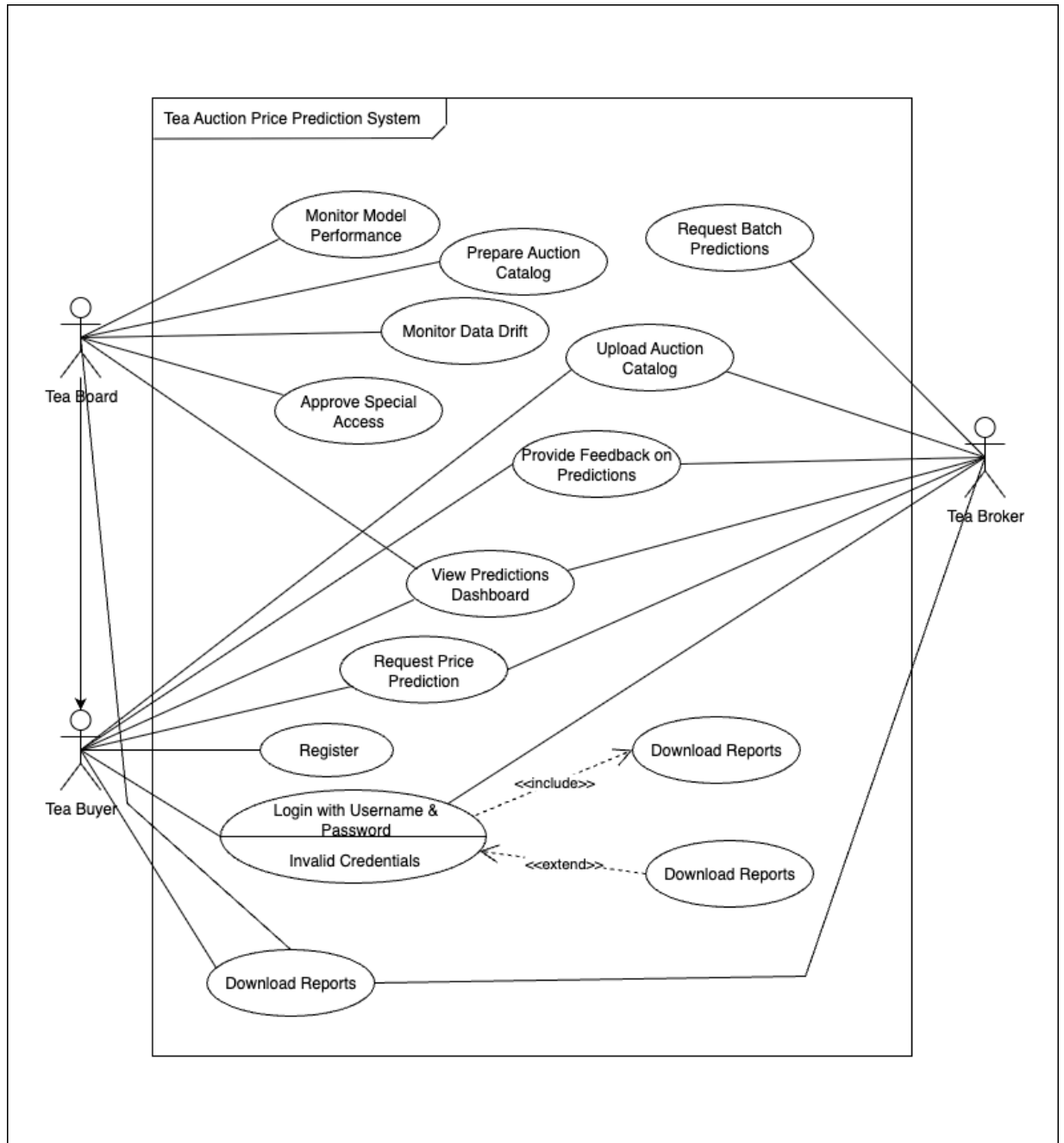


Figure 2 Data-Driven Tea Auction Price Prediction Use Case Diagram

3.7 Work Breakdown Structure

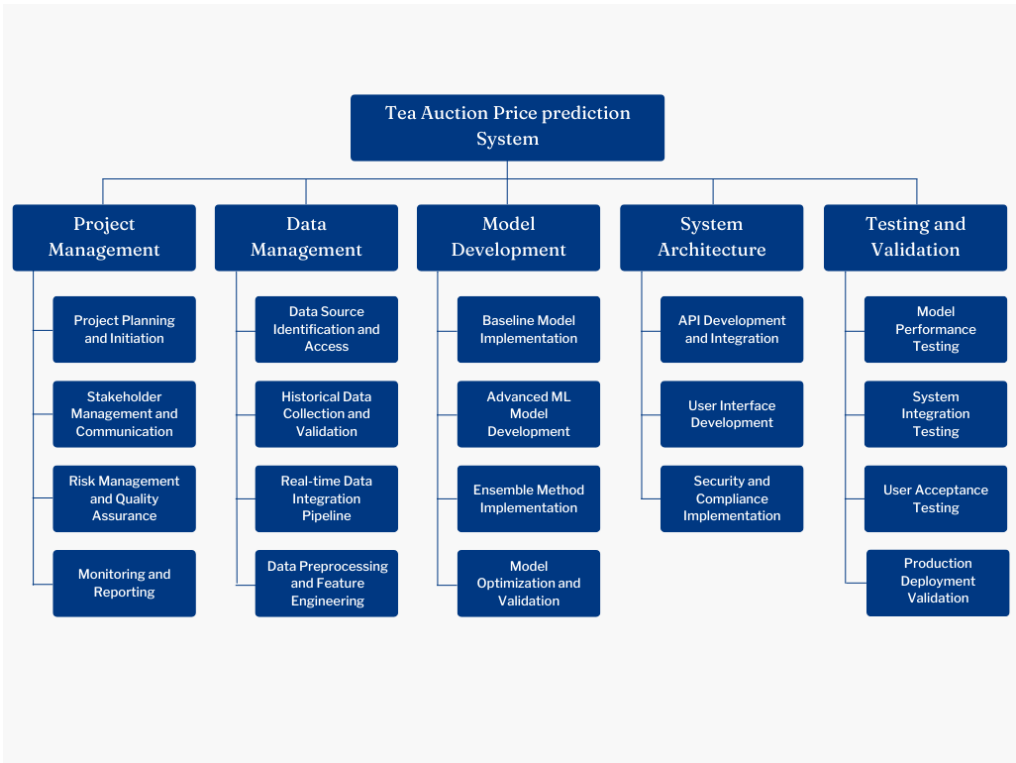


Figure 3 Data-Driven Tea Auction Price Prediction Use Case Diagram

3.8 Gantt Chart

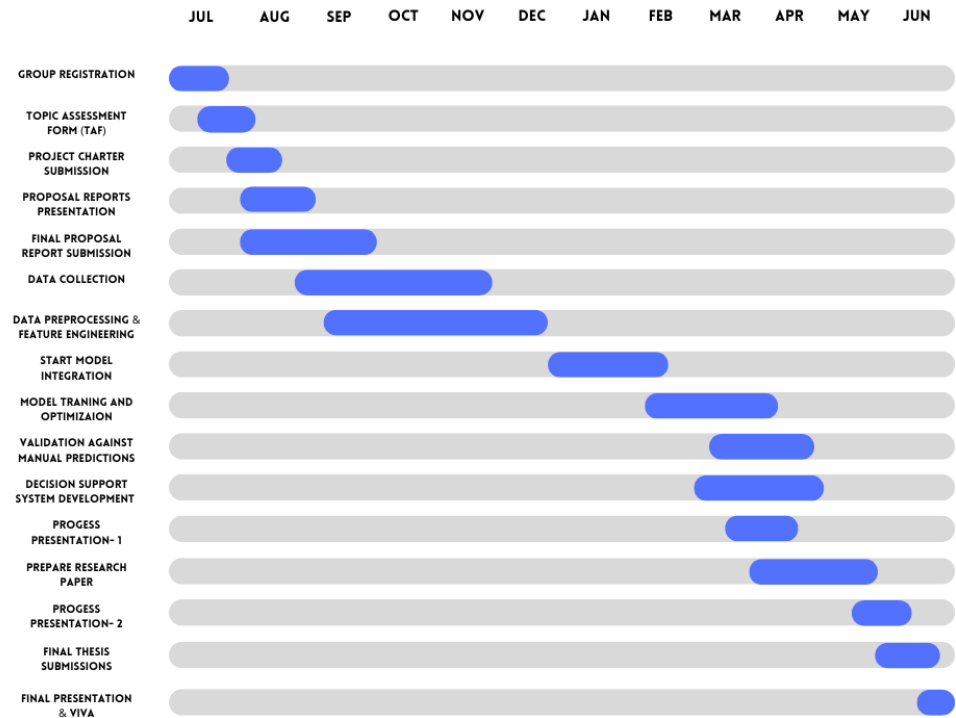


Figure 4 Data-Driven Tea Auction Price Prediction Gantt Chart

3.9 Project Requirements

3.9.1 System Requirements

Functional Requirements

- The intended task of the system is to make auction prices predictions of the individual tea lots as close and accurate as possible or better than the XGBoost benchmark models.
- The system should give price estimates on the seven agro-climatic regions with each region having measurements of the accuracy.
- Meteorological inputs such as rainfall, temperature and humidity need to be implied in the system.
- The system should be able to provide web-based dashboards to address the needs of the producers, broker and buyers.
- All the predictions and interactions with the system must be audited and kept in logs.

Non-Functional Requirements

- The system should be in a position to provide individual prediction responses averagely of 200 milliseconds.
- The system should give concise instructions and feedback which should also be free of ambiguity and consistent with pertinent data protection laws.
- The system will also use the access role-based control authentication.

3.9.2 User Requirements

- Buyers need real-time price predictions to optimize their bidding strategy.
- Buyers also need quality-price correlation analysis for value assessment.
- Brokers need comprehensive market analysis tools for advisory services to their clients.
- Producers need accurate price forecasts to plan their production.

4. Description of Personal and Facilities

Final year undergraduate student in Information Technology with already have some experience in machine learning and Data analytics. In the past, experience includes applying data preprocessing and feature engineering techniques [2], [5]. Both academically and professionally there is strong understanding of machine learning algorithms and their practical applications. Significant experience has been gained in cleaning, transforming, and designing data pipelines to maintain orderly datasets. Being familiar with Python and major machine learning libraries supports performing statistical analysis and identifying patterns in data effectively [2], [5]. Finally the results of the research should be documented and reported along with consistent progress updates throughout the project.

The current study shows that access of the databases offered by the Sri Lanka Tea Board, the administration of which is carried out by the Information Technology and Statistical Division, provides an inclusive set of past auction data and up-to-date market related updates. This information based on the tea industry through the digital auction platform of the industry obtained through CICRA Solutions will record the bidding behaviours, quality evaluation, and pricing patterns since April 2020. Collaborative arrangements with the stakeholders of the tea industry and sector-related knowledge, quality-checking frameworks, and standard datasets needed to develop the model are provided via collaboration agreements. At the same time, the facility has its secure facilities that are specially designed to handle sensitive business information and are in line with the normal confidentiality requirements within the industry [6].

Performance monitoring and analytics platforms are the all-comprehensive tools that provide the accuracy of predictive models, effectiveness of calculations, and scalability of the actual systems. On these platforms, testing protocols such as A/B testing allow comparing different paradigms of machine learning and ensemble configurations[5].

5. Budget and Budget Justification

The overall budget of the entire project Rs. 27,500.00 is a smartly planned and economically sensible investment on four critical components, which cannot be avoided in the process of creating a complete tea auction price prediction system. The fact that Rs. 10,000.00 (36.4%) will be spent on software licenses and tools is also central to providing a secure technical implementation such as scalable deployment of ML models, professional development environments and special APIs. The travel costs of Rs. 7,500.00 (27.3%) support the necessary field research, like visiting tea gardens in different parts of it, viewing events in auction centers and even meeting with stakeholders thereby offering invaluable ground-truth information and empirical confirmation. The allocation of Rs. 5,000.00 (18.2%) to meteorological and external data will ensure access to important weather patterns, global market indices and economic indicator to ensure that accuracy is achieved to the model beyond the local auction parameters. The other Rs. 5,000.00 (18.2%) represent the research materials needed to academic resources, research papers, industry reports, and specialized publications on tea market analysis necessary for comprehensive analysis and professional presentation of results. The balance of the budget substantially maximizes research impact in its use of open-source technologies strategically and yet having enough resources in data quality, technical infrastructure, and actual validation offering unique value to the research community of the Sri Lankan tea industry.

Table 2 System Comparison

Requirements	Cost(Rs)
Software Licenses and Tools	10000.00
Travelling Cost	7500.00
Meteorological and External Data	5000.00
Research Materials and Resources	5000.00
Total	27500.00

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